

CityViews: Benchmarking Satellite and Street-Level Imagery in Urban Vision-Language Models

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Does a second urban view help a vision-language model?

Only when the answer lives *between* the views.

Fusion synergy = accuracy with both views – accuracy of the better single view, averaged over tasks.
 Same 9B model, same questions, same protocol; the two task regimes give opposite answers:

+1.6 pts
 urban-attribute tasks (7)
 land use, road type, building height, density, ...

+46.6 pts
 cross-view tasks (5)
 camera direction, satellite ↔ street matching

1 The problem

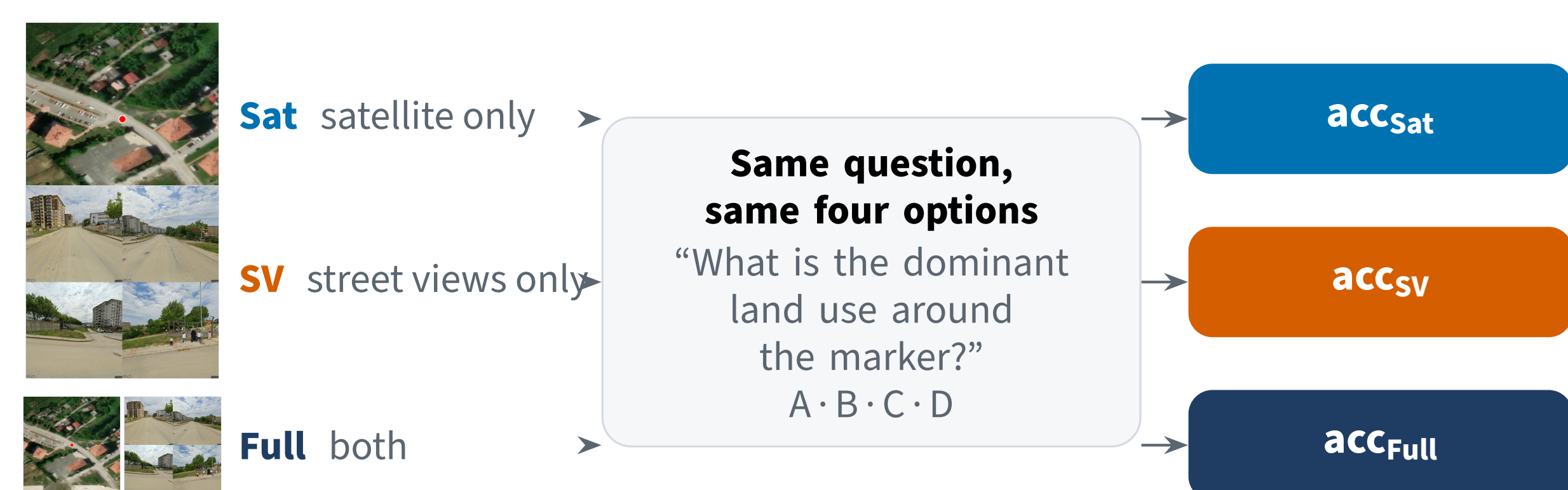
- Satellite tiles and street-level photos are *assumed* to be complementary for urban VLMs, but this is rarely tested directly.
- One aggregate accuracy conflates two very different situations: a question one view could answer alone, and a question that genuinely needs both.
- Street-level coverage is expensive to acquire, license and serve; its *marginal* value should be measured *before* deployment, not after.

2 CityViews at a glance



Each location = one **marked satellite tile** (Esri / NAIP / IGN, sub-metre) + **four road-aligned street views**. Labels from OpenStreetMap tags and geometry, questions from hand-written templates: **no LLM or VLM anywhere in the pipeline**, so CityViews regenerates deterministically from source.

3 The protocol: one question, three inputs

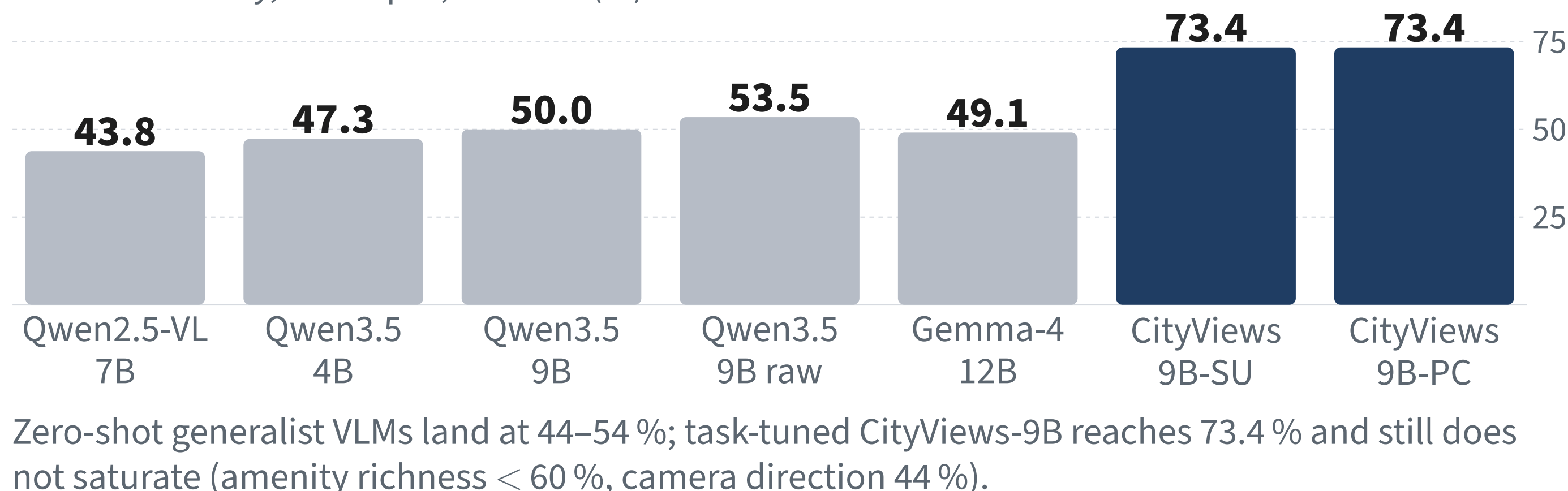


$$\text{synergy} = \text{acc}_{\text{Full}} - \max(\text{acc}_{\text{Sat}}, \text{acc}_{\text{SV}})$$

Synergy within ± 3 pts is read as “no meaningful benefit from the second view” (equivalence band). 4 734 held-out questions; identical text across conditions, so any difference is due to the images.

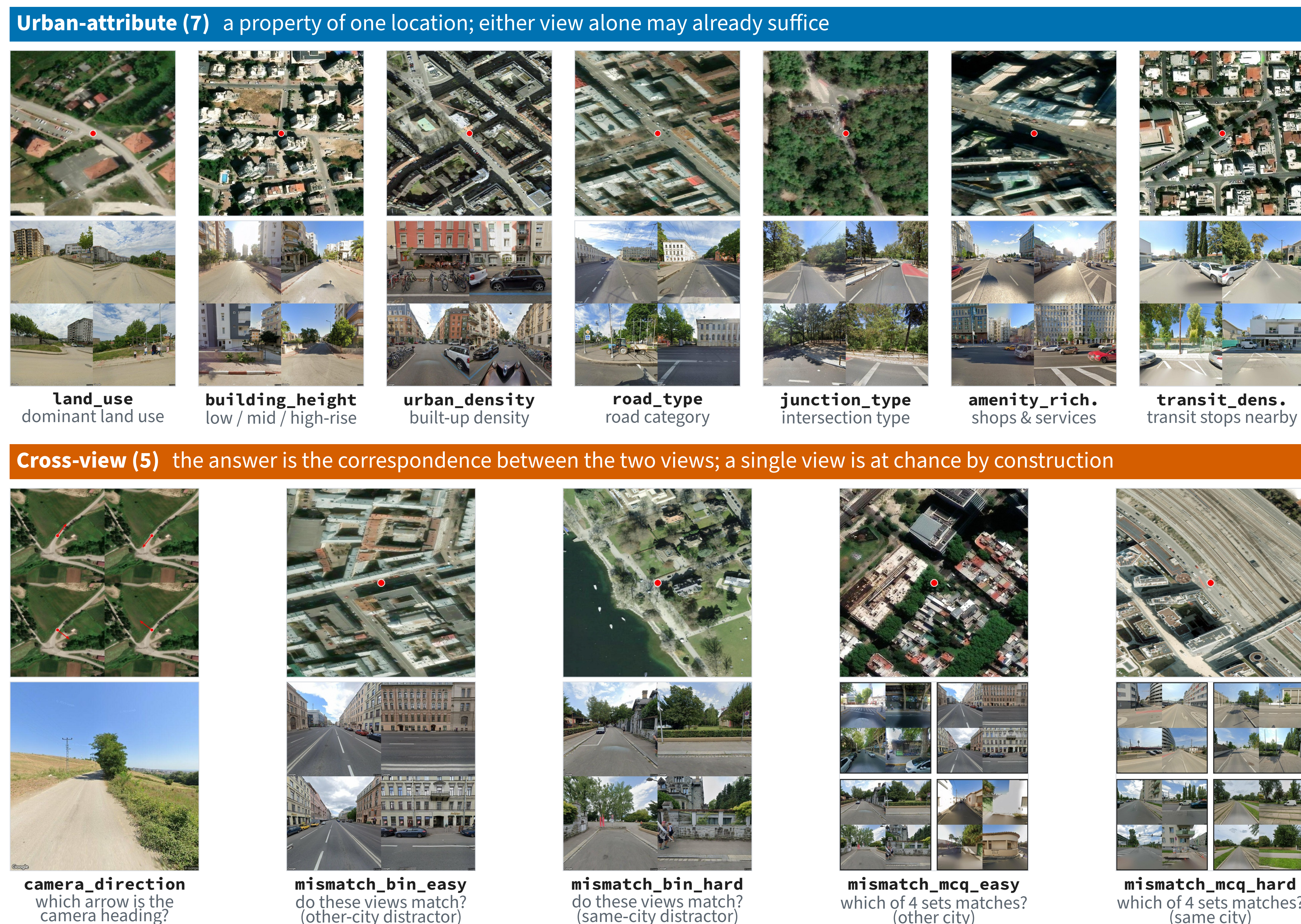
4 Does the benchmark separate models?

overall accuracy, **Full** input, 12 tasks (%)



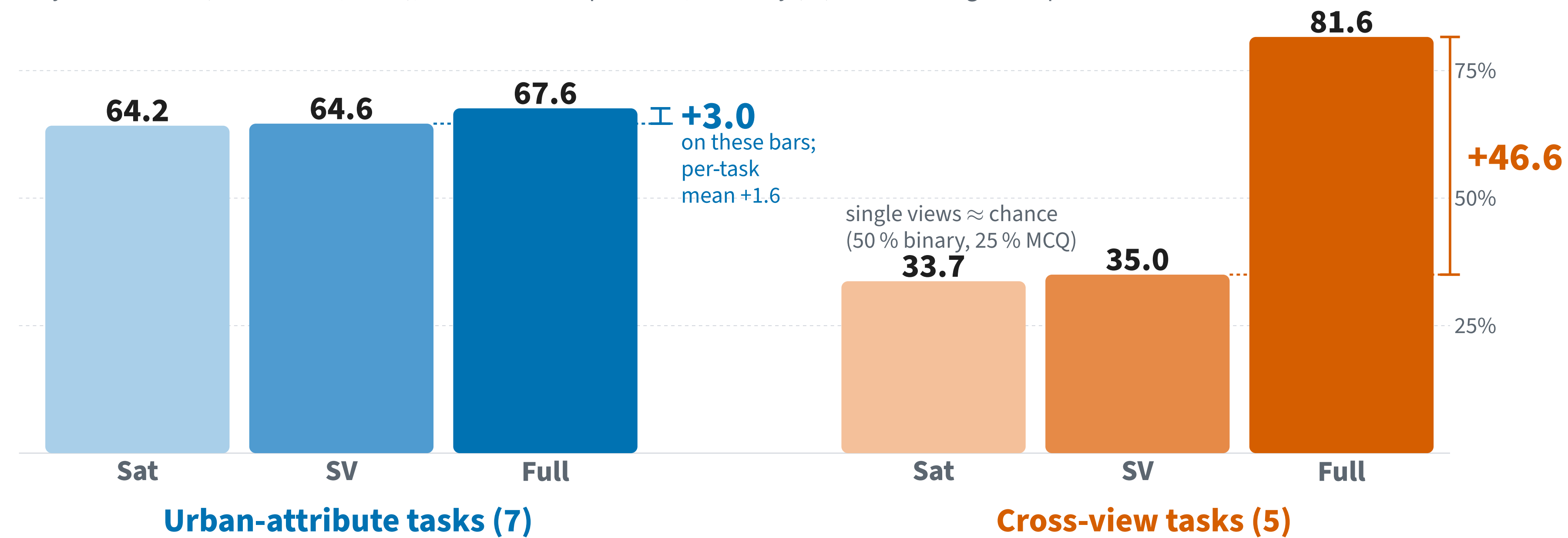
5 Twelve tasks, two regimes

Each cell: marked satellite tile (top) and street view(s) (bottom) for one location. Four-way multiple choice unless noted.

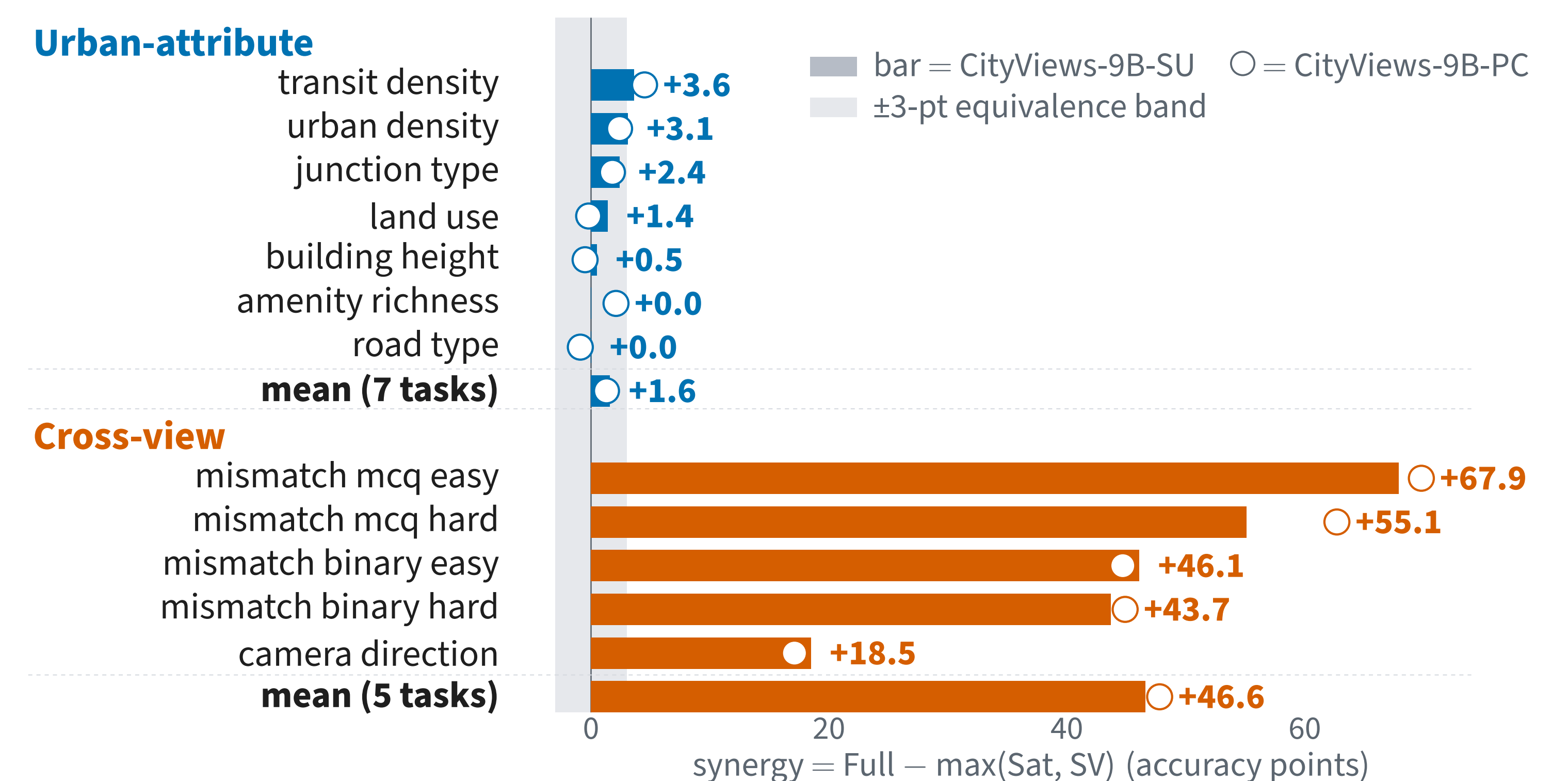


6 Headline: the same model, opposite answers

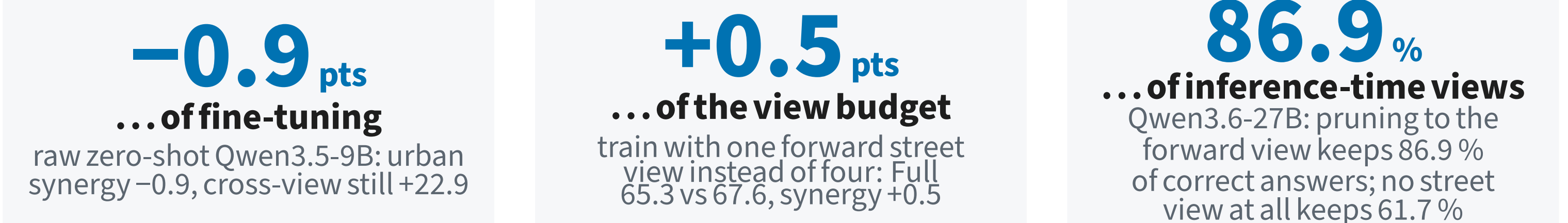
CityViews-9B-SU (Qwen3.5-9B, LoRA), 4 734 held-out questions; accuracy (%) over each regime's questions



7 Per-task fusion synergy



8 The small urban gain is not an artefact



9 Three models, one protocol

Model	Tasks	Sat	SV	Full	Synergy
CityViews-9B-SU	urban (7)	64.2	64.6	67.6	+1.6
	cross-view (5)	33.7	35.0	81.6	+46.6
CityViews-9B-PC	urban (7)	63.9	64.6	66.8	+1.3
	cross-view (5)	34.3	33.7	82.5	+47.8
Qwen3.5-9B, raw	urban (7)	41.3	49.9	49.5	–0.9
	cross-view (5)	35.5	34.4	59.1	+22.9

SU: seen/unseen split (8 held-out cities); PC: per-city split. Accuracy (%) averaged over tasks; Synergy = mean of per-task synergies.

10 Takeaways

- Fusion is **decisive** when the answer is the relationship **between views**, and near-zero when either view already carries the evidence.
- Aggregate both-views accuracy hides this: **report matched single-view baselines**.
- Measure before you collect**: for attribute-style questions, street-level imagery may not pay for itself.

Scope. Behavioural synergy of current VLMs on multiple-choice questions with OSM-certified labels, Europe-weighted; not an information-theoretic claim about the views.

